Implementation of Cy-FFP

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1 Preliminaries

For a system that is axisymmetric, the angular momentum along that axis is conserved based on Noether's theorem [1]. For a rotationally symmetric system around the z-axis, the z-component of angular momentum, denoted by m, is a good quantum number or constant of motion. For this reason, it would be beneficial to decompose the near-field into the angular-momentum eigen-basis $\{e^{im\theta}, m = 0, \pm 1, \pm 2, ...\}$.
The tangential electric field on a near-field monitor plane at some $z = z_{\text{near}}$ is simply given in cylindrical coordinates:

$$\begin{aligned}\mathbf{E}_{\text{Near}} &= \sum_m E_r(r, z_0, m)e^{im\theta}\hat{r} + \sum_m E_\theta(r, z_0, m)e^{im\theta}\hat{\theta} \\ &= \sum_m \mathbf{E}_m^{\text{Near}}(r, z_0)e^{im\theta} \end{aligned} \tag{1}$$

where

$$\mathbf{E}_m^{\text{Near}}(r, z_0) = \left[E_r(r, z_0, m)\hat{r} + E_\theta(r, z_0, m)\hat{\theta}\right] \tag{2}$$

By the way, keep in mind the definition of $\hat{r}$ and $\hat{\theta}$:

$$\hat{r} = \cos\theta\hat{x} + \sin\theta\hat{y} \tag{3}$$

$$\hat{\theta} = -\sin\theta\hat{x} + \cos\theta\hat{y} \tag{4}$$

Conversely, $\hat{x}$ and $\hat{y}$ are given by:

$$\hat{x} = \cos\theta\hat{r} - \sin\theta\hat{\theta} \tag{5}$$

$$\hat{y} = \sin\theta\hat{r} + \cos\theta\hat{\theta} \tag{6}$$

For simplicity, we can let $z_0 = 0$.

2 Vector Cylindrical Harmonics: Eigen-solutions of Vacuum Maxwell's Equations in Cylindrical Coordinates

The idea is to decompose a given near field into eigenmodes of the free-space Maxwell equations in cylindrical coordinates. Each mode is then propagated easily to a desired distance z. The full field at z is then computed by re-summing the propagated eigenmodes. Eigenmodes of cylindrical Maxwell's equations are called vector cylindrical harmonics. There are two types of vector harmonics distinguished by polarization: TM and TE. Both can be found from a scalar Helmholtz equation [**<empty citation>**].

2.1 TM Harmonics

The TM case is given by:

$$\frac{\mathbf{A}}{\mu} = \boldsymbol{\psi} \tag{7}$$

$$\boldsymbol{\psi} = (0, 0, \psi_z) \tag{8}$$

$$\mathbf{B} = \nabla \times \mathbf{A} \tag{9}$$

$$\mathbf{B} = \mu\mathbf{H} \tag{10}$$

$$\mathbf{H} = \frac{1}{\mu}\nabla \times \mathbf{A} \tag{11}$$

$$\nabla \times \mathbf{H} = \epsilon\frac{\partial \mathbf{E}}{\partial t} = -i\omega\epsilon\mathbf{E} \tag{12}$$

$$\nabla \times (\nabla \times \boldsymbol{\psi}) = i\omega\epsilon(-i\omega\psi) \tag{13}$$

$$\nabla^2\psi + \omega^2\mu\epsilon\psi = 0 \tag{14}$$

$$\nabla^2\psi + k^2\psi = 0 \tag{15}$$

which is the Helmholtz equation[2]. Note that as usual:

$$k = \omega\sqrt{\mu\epsilon} = \frac{2\pi n}{\lambda} \tag{16}$$

In cylindrical coordinates $(\mathbf{r}, \theta, z)$:

$$\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial\psi}{\partial r}\right) + \frac{1}{r^2}\frac{\partial^2\psi}{\partial\theta^2} + \frac{\partial^2\psi}{\partial z^2} + k^2\psi = 0 \tag{17}$$

We can solve it by the method of separation of variables[2]:

$$\psi = R(r)\Theta(\theta)Z(z) \tag{18}$$

Substituting into the Helmholtz equation:

$$\frac{1}{rR}\frac{d}{dr}\left(r\frac{dR}{dr}\right) + \frac{1}{r^2\Theta}\frac{d^2\Theta}{d\theta^2} + \frac{1}{Z}\frac{d^2Z}{dz^2} + k^2 = 0 \tag{19}$$

Separating the z-dependence:

$$\frac{1}{Z}\frac{d^2Z}{dz^2} = -k_z^2 \tag{20}$$

Substituting k_z back:

$$\frac{1}{rR}\left(\frac{d}{dr}\left(r\frac{dR}{dr}\right)\right) + \frac{1}{r^2\Theta}\frac{d^2\Theta}{d\theta^2} + k^2 - k_z^2 = 0 \tag{21}$$

Now we can define k_r as:

$$k_r^2 = k^2 - k_z^2 \tag{22}$$

Separating the θ-dependence:

$$\frac{1}{\Theta}\frac{d^2\Theta}{d\theta^2} = -m^2 \tag{23}$$

The radial equation becomes:

$$\frac{d}{dr}\left(r\frac{dR}{dr}\right) + \left(k_r^2 r - \frac{m^2}{r}\right)R = 0 \tag{24}$$

which is the radial form of the differential Bessel equation [3]. The solution for ψ is:

$$\psi = J_m(k_r r)\exp(im\theta)\exp(ik_z z) \tag{25}$$

We can now proceed to find the fields. The magnetic field is given by $\mathbf{H} = \nabla \times \boldsymbol{\psi}$:

$$\mathbf{H} = \nabla \times \boldsymbol{\psi} = \hat{r}\left(\frac{1}{r}\frac{\partial\psi_z}{\partial\theta} - \frac{\partial\psi_\theta}{\partial z}\right) + \hat{\theta}\left(\frac{\partial\psi_r}{\partial z} - \frac{\partial\psi_z}{\partial r}\right) + \hat{z}\frac{1}{r}\left(\frac{\partial(r\psi_\theta)}{\partial r} - \frac{\partial\psi_r}{\partial\theta}\right) \tag{26}$$

For $\boldsymbol{\psi} = (0, 0, \psi_z)$ magnetic field becomes:

$$\mathbf{H} = \begin{pmatrix} \frac{1}{r}\frac{\partial}{\partial\theta}\psi \\ -\frac{\partial}{\partial r}\psi \\ 0 \end{pmatrix} \tag{27}$$

Plugging in (25) gives:

$$\mathbf{H} = \left(\frac{im}{r}J_m\hat{r} - \frac{\partial J_m}{\partial r}\hat{\theta}\right)\exp(im\theta)\exp(-ik_z z) \tag{28}$$

The electric field $\mathbf{E}$ is found from Maxwell's equation $\nabla \times \mathbf{H} = i\omega\epsilon\mathbf{E}$, so $\mathbf{E} = \frac{1}{i\omega\epsilon}\nabla \times \mathbf{H}$:

$$i\omega\epsilon\mathbf{E} = \nabla \times \mathbf{H} = \hat{r}\left(\frac{1}{r}\frac{\partial H_z}{\partial\theta} - \frac{\partial H_\theta}{\partial z}\right) + \hat{\theta}\left(\frac{\partial H_r}{\partial z} - \frac{\partial H_z}{\partial r}\right) + \hat{z}\frac{1}{r}\left(\frac{\partial(rH_\theta)}{\partial r} - \frac{\partial H_r}{\partial\theta}\right) \tag{29}$$

Since $H_z = 0$, the components of $\mathbf{E}$ are:

$$E_r = -\frac{1}{i\omega\epsilon}\frac{\partial H_\theta}{\partial z} = \frac{1}{i\omega\epsilon}\frac{\partial}{\partial z}\left(\frac{\partial\psi}{\partial r}\right) = -\frac{i}{\omega\epsilon}\frac{\partial^2\psi}{\partial z\partial r} \tag{30}$$

$$E_\theta = \frac{1}{i\omega\epsilon}\frac{\partial H_r}{\partial z} = \frac{1}{i\omega\epsilon}\frac{\partial}{\partial z}\left(\frac{1}{r}\frac{\partial\psi}{\partial\theta}\right) = -\frac{i}{\omega\epsilon}\frac{1}{r}\frac{\partial^2\psi}{\partial z\partial\theta} \tag{31}$$

$$E_z = \frac{1}{i\omega\epsilon}\frac{1}{r}\left(\frac{\partial(rH_\theta)}{\partial r} - \frac{\partial H_r}{\partial\theta}\right) = -\frac{i}{\omega\epsilon}\left(\frac{1}{r}\frac{\partial}{\partial r}\left(-r\frac{\partial\psi}{\partial r}\right) - \frac{1}{r^2}\frac{\partial^2\psi}{\partial\theta^2}\right) \tag{32}$$

Using the Helmholtz equation in cylindrical coordinates $+\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial\psi}{\partial r}\right) + \frac{1}{r^2}\frac{\partial^2\psi}{\partial\theta^2} + \frac{\partial^2\psi}{\partial z^2} + k^2\psi = 0$:

$$\begin{aligned} E_z &= \left(\frac{i}{\omega\epsilon}\right)\left(\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial\psi}{\partial r}\right) + \frac{1}{r^2}\frac{\partial^2\psi}{\partial\theta^2}\right) \\ &= \left(\frac{i}{\omega\epsilon}\right)\left(\nabla^2_{\text{cylindrical}}\psi - \frac{\partial^2\psi}{\partial z^2}\right) \\ &= \left(\frac{i}{\omega\epsilon}\right)\left(k^2\psi - \frac{\partial^2\psi}{\partial z^2}\right) \end{aligned} \tag{33}$$

So electric field becomes:

$$\mathbf{E}^{\text{TM}}_{m,k_r} = \sqrt{\frac{\mu}{\epsilon}}\left(-\frac{k_z}{k}\frac{\partial J_m(k_r r)}{\partial r}\hat{r} - i\frac{mk_z}{kr}J_m(k_r r)\hat{\theta} + i\frac{k_r^2}{k}J_m(k_r r)\hat{z}\right)\exp(im\theta)\exp(-ik_z z) \tag{34}$$

2.2 TE Harmonics

Similarly for the case of TE:

$$\frac{\mathbf{F}}{\epsilon} = \boldsymbol{\psi} \tag{35}$$

$$\boldsymbol{\psi} = (0, 0, \psi_z) \tag{36}$$

$$\nabla \cdot \mathbf{D} = 0 \tag{37}$$

$$\mathbf{D} = \nabla \times \mathbf{F} \tag{38}$$

$$\mathbf{D} = -\epsilon\mathbf{E} \tag{39}$$

From $\mathbf{D} = \nabla \times \mathbf{F} = \nabla \times (\epsilon\boldsymbol{\psi})$ and $\mathbf{E} = -\frac{1}{\epsilon}\mathbf{D}$:

$$\mathbf{E} = -\frac{1}{\epsilon}\nabla \times \mathbf{F} = -\nabla \times \boldsymbol{\psi} \tag{40}$$

$$\mathbf{E} = -\nabla \times \boldsymbol{\psi} = -\left[\hat{r}\left(\frac{1}{r}\frac{\partial\psi_z}{\partial\theta} - \frac{\partial\psi_\theta}{\partial z}\right) + \hat{\theta}\left(\frac{\partial\psi_r}{\partial z} - \frac{\partial\psi_z}{\partial r}\right) + \hat{z}\frac{1}{r}\left(\frac{\partial(r\psi_\theta)}{\partial r} - \frac{\partial\psi_r}{\partial\theta}\right)\right] \tag{41}$$

Given $\boldsymbol{\psi} = (0, 0, \psi_z)$ and letting $\psi = \psi_z$:

$$\mathbf{E} = \begin{pmatrix} -\frac{1}{r}\frac{\partial}{\partial\theta}\psi \\ \frac{\partial}{\partial r}\psi \\ 0 \end{pmatrix} \tag{42}$$

Inserting ψ, the electric field becomes:

$$\mathbf{E}^{\text{TE}}_{m,k_r}(r,\theta,z) = -\left(\hat{r}\frac{im}{r}J_m - \hat{\theta}\frac{\partial J_m}{\partial r}\right)\exp(im\theta)\exp(ik_z z) \tag{43}$$

Remember $k_z = k_z(k_r) = \sqrt{k^2 - k_r^2}$.
The magnetic field of TE case can be found similar to TM case.

3 Near to Far Propagation

Now that we have found the cylindrical harmonics, near-field can be expressed in the cylindrical eigenbasis by integrating over all possible radial wavevectors and summing over the angular momentum m;

$$\mathbf{E}_{\text{Near}}(r,\theta) = \mathbf{E}(r,\theta,z=z_0) = \sum_m \int dk_r \left[A_m^{\text{TE}}(k_r)\mathbf{E}_{m,k_r}^{\text{TE}}(r,\theta,z_0) + A_m^{\text{TM}}(k_r)\mathbf{E}_{m,k_r}^{\text{TM}}(r,\theta,z_0)\right] \tag{44}$$

where $\mathbf{E}_{m,k_r}^{\text{TE}}$ and $\mathbf{E}_{m,k_r}^{\text{TM}}$ show the cylindricl harmonics for TE and TM case. Then at $z = z_1$, we will have:

$$\mathbf{E}_{\text{Far}}(r,\theta) = \mathbf{E}(r,\theta,z=z_1) = \sum_m \int dk_r \left[A_m^{\text{TE}}(k_r)\mathbf{E}_{m,k_r}^{\text{TE}}(r,\theta,z_1) + A_m^{\text{TM}}(k_r)\mathbf{E}_{m,k_r}^{\text{TM}}(r,\theta,z_1)\right] \tag{45}$$

Note that at $z = z_1$, the eigenmodes have the propagation factor: $e^{ik_z z_1}$ (whereas at $z = z_0 = 0$ there is no such factor). Here, it is important to remember $k_z = k_z(k_r) = \sqrt{k^2 - k_r^2}$. Therefore, the k_r-integration in Eq. 45 must include $e^{ik_z z}$ (which cannot be taken outside of the k_r-integration).
In order to determine the coefficients $A_m^{\text{TE}}(k_r)$ and $A_m^{\text{TM}}(k_r)$, we need to project $\mathbf{E}_{\text{Near}}$ onto the (m',k_r') TE and TM eigenmodes seperately. Let's start with the TE coefficient. In the right hand side the second term vanishes due to the orthogonality of the TE and TM modes. It is also obvious that only the terms $m = m'$ will survive when integrated over θ. So we can just look at the projection onto $\mathbf{E}_{m,k_r'}^{\text{TE}}$:

$$\begin{aligned}
\mathbf{E}_{m,k_r}^{\text{TE}} \cdot \mathbf{E}_{m,k_r'}^{\text{TE}^*} &= -\left(\hat{r}\frac{im}{r}J_m(k_r r) - \hat{\theta}\frac{\partial J_m(k_r r)}{\partial r}\right)\exp(im\theta) \\
&\quad \cdot -\left(-\hat{r}\frac{im}{r}J_m(k_r' r) - \hat{\theta}\frac{\partial J_m(k_r' r)}{\partial r}\right)\exp(-im\theta) && (46)\\
&= \left(\frac{m^2}{r^2}J_m(k_r r)J_m(k_r' r) + k_r k_r'\frac{\partial J_m(k_r r)}{\partial k_r r}\frac{\partial J_m(k_r' r)}{\partial k_r' r}\right) && (47)
\end{aligned}$$

Note that due to the orthogonality of different modes [2], m and m' need to be equal.
The following recurrence relations are tabulated in the NIST Digital Library of Mathematical Functions [4]:

$$J_{m+1}(x) + J_{m-1}(x) = \frac{2m}{x}J_m(x) \tag{48}$$

$$J_{m-1}(x) - J_{m+1}(x) = 2\frac{dJ_m(x)}{dx} \tag{49}$$

Substituting gives:

$$\begin{aligned}
\mathbf{E}_{m,k_r}^{\text{TE}} \cdot \mathbf{E}_{m,k_r'}^{\text{TE}*} &= \left(\frac{m^2}{r^2}\cdot\frac{k_r k_r' r^2}{4m^2}\right)\big(J_{m+1}(k_r r) + J_{m-1}(k_r r)\big)\big(J_{m+1}(k_r' r) + J_{m-1}(k_r' r)\big) \\
&\quad + \frac{k_r k_r'}{4}\big(-J_{m+1}(k_r r) + J_{m-1}(k_r r)\big)\big(-J_{m+1}(k_r' r) + J_{m-1}(k_r' r)\big) && (50)\\
&= \left(\frac{k_r k_r'}{2}\right)\left[J_{m+1}(k_r r)J_{m+1}(k_r' r) + J_{m-1}(k_r r)J_{m-1}(k_r' r)\right] && (51)
\end{aligned}$$

The orthogonality relation for Bessel functions of the first kind is given by [5]:

$$\int_0^\infty r J_m(k_r r)J_m(k_r' r)dr = \frac{\delta(k_r - k_r')}{k_r} \tag{52}$$

Thus, using Eq. 2, we obtain:

$$\iint \mathbf{E}_{\text{Near}} \cdot \mathbf{E}_{m,k_r}^{\text{TE}^*}\, r dr d\theta = \iint \left(\sum_{m'} \mathbf{E}_{m'}^{\text{Near}} e^{im'\theta}\right) \cdot \left(-\left(\hat{r}\frac{im}{r}J_m - \hat{\theta}\frac{\partial J_m}{\partial r}\right)^* \exp(-im\theta)\right)\, r dr d\theta = 2\pi k_r A_m^{\text{TE}} \tag{53}$$

Integrating out the θ, A_m^{TE} coefficient becomes:

$$\begin{aligned}
A_m^{\text{TE}}(k_r) &= \frac{1}{k_r}\int \mathbf{E}_m^{\text{Near}}(r) \cdot \left(imk_r\frac{J_m(k_r r)}{k_r r}\,\hat{r} + k_r\frac{\partial J_m(k_r r)}{\partial k_r r}\,\hat{\theta}\right)\, r dr && (54)\\
&= \int \mathbf{E}_m^{\text{Near}}(r) \cdot \left(im\frac{J_m(k_r r)}{k_r r}\,\hat{r} + \frac{\partial J_m(k_r r)}{\partial k_r r}\,\hat{\theta}\right)\, r dr && (55)\\
&= \frac{1}{2}\int \mathbf{E}_m^{\text{Near}}(r) \cdot \Big(\big[iJ_{m+1}(k_r r) + iJ_{m-1}(k_r r)\big]\,\hat{r} + \big[J_{m-1}(k_r r) - J_{m+1}(k_r r)\big]\,\hat{\theta}\Big)\, r dr && (56)
\end{aligned}$$

Do the same thing for A_m^{TM} that I did for A_m^{TE}. XXXXXXXXXX
Using $A_m^{\mathrm{TE}}(k_r)$ and $A_m^{\mathrm{TM}}(k_r)$, the far field can be explicitly written as:

$$XXX \tag{57}$$

Using the equations (**??**),(**??**) and (**??**) farfield can be obtained using 6 Hankel Transforms for each m.

4 Differentiable Fast Hankel Transform

5 Example validations

5.1 Ideal lens under normal incidence

We will validate our code on a few near-field examples. For example, we know the x-polarized electric field of an ideal lens (under normal incidence) and we put it into the form of Eq. 2 in order to find multiple $\mathbf{E}_m^{\mathrm{Near}}$'s that can be fed directly into our FFP formulae:

$$\begin{aligned}
\mathbf{E} &= \exp -i\frac{2\pi}{\lambda}\left(\sqrt{x^2+y^2+f^2}-f\right)\hat{x} && (58)\\
&= \exp -i\frac{2\pi}{\lambda}\left(\sqrt{r^2+f^2}-f\right)\hat{x} && (59)\\
&= u(r;f,\lambda)\ \left(\cos\theta\hat{r}-\sin\theta\hat{\theta}\right) && (60)\\
&= \frac{u(r)}{2}\left((e^{i\theta}+e^{-i\theta})\hat{r}-(e^{i\theta}-e^{-i\theta})\hat{\theta}\right) && (61)\\
&= \frac{u(r)}{2}e^{i\theta}\hat{r}-\frac{u(r)}{2}e^{i\theta}\hat{\theta}+\frac{u(r)}{2}e^{-i\theta}\hat{r}+\frac{u(r)}{2}e^{-i\theta}\hat{\theta} && (62)\\
&= \left(\frac{u(r)}{2}\hat{r}-\frac{u(r)}{2}\hat{\theta}\right)e^{i\theta}+\left(\frac{u(r)}{2}\hat{r}+\frac{u(r)}{2}\hat{\theta}\right)e^{-i\theta} && (63)\\
&= \mathbf{E}_{+1}^{\mathrm{Near}}(r)\ e^{i\theta}+\mathbf{E}_{-1}^{\mathrm{Near}}(r)\ e^{-i\theta} && (64)
\end{aligned}$$

Side note: For planar current sheets that produce linearly-polarized, normal-incident plane-waves, the excited field satisfies:

$$\begin{aligned}
E_{-1,r}^{\mathrm{Near}} &= E_{1,r}^{\mathrm{Near}} && (65)\\
E_{-1,\theta}^{\mathrm{Near}} &= -E_{1,\theta}^{\mathrm{Near}} && (66)
\end{aligned}$$

even for arbitrary $\mathbf{E}_{\mathrm{Near}}$. This means that you normally shouldn't have to do FDTD for $m=-1$, which can be read off from the positive $m=+1$. (Proof can be found in the notes for cylindrical solver implementation. See also the appendix.)

5.2 Ideal lens under oblique incidence

For a general oblique incident angle α, we can generalize the ideal lens field as follows:

$$\begin{aligned}
\sum_m \mathbf{E}_m^{\mathrm{Near}}(r)e^{im\theta} &= u(r,\theta;f,\lambda,\alpha)\ \left(\cos\theta\hat{r}-\sin\theta\hat{\theta}\right) && (67)\\
u(r,\theta;f,\lambda,\alpha) &= XXX && (68)
\end{aligned}$$

From Eq. 67, we must find sufficiently many $\mathbf{E}_m^{\mathrm{Near}}(r)$'s to propagate to the far field.

5.3 Flat metalens with paraxial shift invariance

Rather than testing on ideal near fields, we also tested our code on the transmitted near fields of optimized flat metalenses. The transmitted electric field in the local metalens model is generally given by:

$$\mathbf{E}_{\mathrm{trans}}(x,y) = t(x,y;\lambda,\alpha)\mathbf{E}_{\mathrm{inc}}(x,y) \tag{69}$$

Here, the spatially-varying transmission function $t(x,y;\lambda,\alpha)$ can be computed in RCWA using locally periodic approximation (LPA).
Consider a metalens which has been optimized to focus under normal incidence $\alpha=0$. Virtually every thin flat lens obeys paraxial shift invariance, which means that if the thin lens focuses perfectly at $r=0$ for normal

incidence $\alpha = 0$, it must also focus pretty nicely at $r = f\tan(\alpha)$ for small angles $\alpha \approx 0°$. We will test our code on this: given a normal-incidence-optimized metalens design, we will find the metalens PSF at a small oblique incident angle $\alpha = 5°$.
We re-write the transmitted field model in cylindrical coordinates:

$$\mathbf{E}_{\text{trans}}(r,\theta) = t(r,\theta;\lambda,\alpha)\exp\left(i(k\sin\alpha)(r\cos\theta)\right)\hat{x} \tag{70}$$

Since the metalens is structurally axisymmetric, we have:

$$\mathbf{E}_{\text{trans}}(r,\theta) = t(r;\lambda,\alpha)\exp\left(i(k\sin\alpha)(r\cos\theta)\right)\hat{x} \tag{71}$$

Since $\mathbf{E}_{\text{trans}}$ is simply the near field to be propagated, we must find sufficiently many $\mathbf{E}_m^{\text{Near}}(r)$'s such that

$$\sum_m \mathbf{E}_m^{\text{Near}}(r)e^{im\theta} = t(r;\lambda,\alpha)\exp\left(i(k\sin\alpha)(r\cos\theta)\right)\ \left(\cos\theta\hat{r} - \sin\theta\hat{\theta}\right) \tag{72}$$

6 Scalar version

For sanity check, let's also consider the scalar version of all this. The scalar cylindrical harmonic is simply:

$$\psi_{m,k_r}(r,\theta,z) = J_m(k_r r)e^{im\theta}e^{ik_z z} \tag{73}$$

Any scalar near field is given by:

$$u(r,\theta,z_0=0) = \sum_m \int a_m(kr)J_m(k_r r)e^{im\theta}\ dk_r \tag{74}$$

Then

$$\int\int J_{m'}(k_r' r)e^{-im'\theta}u(r,\theta)\ rdrd\theta = \sum_m \int\int\int J_{m'}(k_r' r)e^{-im'\theta}a_m(kr)J_m(k_r r)e^{im\theta}\ rdrd\theta dk_r \tag{75}$$

$$= \sum_m \int 2\pi\delta_{m,m'}\frac{\delta(k_r - k_r')}{k_r}a_m(k_r)\ dk_r \tag{76}$$

$$= \frac{2\pi}{k_r'}a_{m'}(k_r') \tag{77}$$

So

$$a_m(k_r) = \frac{k_r}{2\pi}\int\int J_m(k_r r)e^{-im\theta}u(r,\theta)\ rdrd\theta \tag{78}$$

We must always write $u(r,\theta)$ as

$$u(r,\theta) = \sum_m u_m(r)e^{im\theta} \tag{79}$$

We then have

$$a_m(k_r) = k_r\int J_m(k_r r)u_m(r)\ rdr \tag{80}$$

The far field is then

$$u(r,\theta,z_1) = \sum_m \left(\int\left[\int J_m(k_r\rho)u_m(\rho)\ \rho d\rho\right]J_m(k_r r)e^{ik_z z_1}\ k_r dk_r\right)e^{im\theta}, \quad k_z = \sqrt{k^2 - k_r^2} \tag{81}$$

Consider the oblique lens profile:

$$u(r,\theta) = \exp\left(-ik\left[\sqrt{(r\cos\theta - f\tan\alpha)^2 + r^2\sin^2\theta + f^2} - f\right]\right) \tag{82}$$

$$= \exp\left(-ik\left[\sqrt{r^2+f^2} - f - \frac{fr\alpha}{\sqrt{r^2+f^2}}\cos\theta\right]\right), \quad \text{for } \alpha \approx 0 \text{ radian} \tag{83}$$

We need to find

$$u_m(r) = \int_0^{2\pi} \exp\left(-ik\Big[\sqrt{r^2+f^2} - f - \frac{fr\alpha}{\sqrt{r^2+f^2}}\cos\theta\Big]\right) e^{-im\theta}\ d\theta \tag{84}$$

$$= \exp\left(-ik\Big[\sqrt{r^2+f^2} - f\Big]\right) \int_0^{2\pi} \exp\left(-\frac{ikfr\alpha}{\sqrt{r^2+f^2}}\cos\theta\right) e^{-im\theta}\ d\theta \tag{85}$$

$$= \exp\left(-ik\Big[\sqrt{r^2+f^2} - f\Big]\right) \int_0^{2\pi} \exp\left(-iA\cos\theta - im\theta\right)\ d\theta \tag{86}$$

Note that

$$A = \frac{kfr\alpha}{\sqrt{r^2+f^2}} \tag{87}$$

We have the identity of the Bessel's first integral:

$$\int_0^{2\pi} \exp\Big[iA\cos\theta - im\theta\Big]\ d\theta = 2\pi i^m J_m(A). \tag{88}$$

Therefore,

$$u_m(r) = 2\pi i^m \exp\left(-ik\Big[\sqrt{r^2+f^2} - f\Big]\right) J_m\left(\frac{kfr\alpha}{\sqrt{r^2+f^2}}\right) \tag{89}$$

Substitute these $u_m(r)$'s into:

$$u(r,\theta,z_1) = \sum_m \left(\int \Big[\int J_m(k_r\rho)u_m(\rho)\ \rho d\rho\Big] J_m(k_r r) e^{ik_z z_1}\ k_r dk_r\right) e^{im\theta}, \quad k_z = \sqrt{k^2 - k_r^2} \tag{90}$$

6.1 Local transmission model under oblique incidence

Consider the local transmission model:

$$u(r,\theta) = t(r;\alpha)u_{\text{inc}}(r,\theta) = t(r;\alpha)e^{ik_x x} \tag{91}$$

$$= t(r;\alpha)\exp\left(ik\sin\alpha\ r\cos\theta\right) \tag{92}$$

Suppose the metalens is angle-insensitive:

$$t(r;\alpha) = t(r;\alpha=0) \tag{93}$$

$$= \exp\left(-ik\Big[\sqrt{r^2+f^2} - f\Big]\right) \tag{94}$$

Then

$$u(r,\theta) = \exp\left(-ik\Big[\sqrt{r^2+f^2} - f\Big]\right)\exp\left(ik\sin\alpha\ r\cos\theta\right) \tag{95}$$

We need to find

$$u_m(r) = \exp\left(-ik\Big[\sqrt{r^2+f^2} - f\Big]\right) \int_0^{2\pi} \exp\left(ik\sin\alpha\ r\cos\theta\right) e^{-im\theta}\ d\theta \tag{96}$$

$$= \exp\left(-ik\Big[\sqrt{r^2+f^2} - f\Big]\right) \int_0^{2\pi} \exp\left(-iA\cos\theta - im\theta\right)\ d\theta, \tag{97}$$

$$A = -k\sin\alpha\ r \tag{98}$$

Therefore,

$$u_m(r) = 2\pi i^m \exp\left(-ik\Big[\sqrt{r^2+f^2} - f\Big]\right) J_m\left(-k\sin\alpha\ r\right) \tag{99}$$

$$= 2\pi(-i)^m \exp\left(-ik\Big[\sqrt{r^2+f^2} - f\Big]\right) J_m\left(k\sin\alpha\ r\right) \tag{100}$$

Substitute these $u_m(r)$'s into:

$$u(r,\theta,z_1) = \sum_m \left(\int \Big[\int J_m(k_r\rho)u_m(\rho)\ \rho d\rho\Big] J_m(k_r r) e^{ik_z z_1}\ k_r dk_r\right) e^{im\theta}, \quad k_z = \sqrt{k^2 - k_r^2} \tag{101}$$

Appendix: cylindrical expansion of plane waves

For an oblique incident angle characterized by (α, ϕ), the x-polarized current sheet is given by:

$$\mathbf{J}_m(r) = P_m(r)\ \hat{r} - Q_m(r)\ \hat{\theta} \tag{102}$$

where

$$P_m(r) = \pi\Big[\ i^{m+1}e^{-i(m+1)\phi}\ J_{m+1}(kr\sin\alpha) +\ i^{m-1}e^{-i(m-1)\phi}\ J_{m-1}(kr\sin\alpha)\Big] \tag{103}$$

$$Q_m(r) = i\pi\Big[\ i^{m+1}e^{-i(m+1)\phi}\ J_{m+1}(kr\sin\alpha) -\ i^{m-1}e^{-i(m-1)\phi}\ J_{m-1}(kr\sin\alpha)\Big] \tag{104}$$

For $\phi = 0$, we can prove that P_m is even in m and Q_m is odd in m.

$$\begin{aligned}
P_{-m}(r) &= \pi\Big[\ i^{-m+1}\ J_{-m+1}(kr\sin\alpha) +\ i^{-m-1}\ J_{-m-1}(kr\sin\alpha)\Big] && (105)\\
&= \pi\Big[\ i^{-(m-1)}(-1)^{m-1}\ J_{m-1}(kr\sin\alpha) +\ i^{-(m+1)}(-1)^{m+1}\ J_{m+1}(kr\sin\alpha)\Big] && (106)\\
&= \pi\Big[\ \frac{1}{i^{m-1}}(-1)^{m-1}\ J_{m-1}(kr\sin\alpha) + \frac{1}{i^{m+1}}(-1)^{m+1}\ J_{m+1}(kr\sin\alpha)\Big] && (107)\\
&= \pi\Big[\ \frac{i^{m-1}}{i^{2(m-1)}}(-1)^{m-1}\ J_{m-1}(kr\sin\alpha) + \frac{i^{m+1}}{i^{2(m+1)}}(-1)^{m+1}\ J_{m+1}(kr\sin\alpha)\Big] && (108)\\
&= \pi\Big[\ i^{m-1}\ J_{m-1}(kr\sin\alpha) + i^{m+1}\ J_{m+1}(kr\sin\alpha)\Big] && (109)\\
&= P_m(r) && (110)
\end{aligned}$$

Likewise, $Q_{-m}(r) = -Q_m(r)$.
Therefore, we have:

$$\mathbf{J}_m = \begin{bmatrix} P_m \\ -Q_m \\ 0 \end{bmatrix} \tag{111}$$

$$\mathbf{J}_{-m} = \begin{bmatrix} P_{-m} \\ -Q_{-m} \\ 0 \end{bmatrix} = \begin{bmatrix} P_m \\ Q_m \\ 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \mathbf{J}_m = \hat{\sigma}\mathbf{J}_m \tag{112}$$

Therefore, we have that

$$\begin{aligned}
\mathcal{D}_m\mathcal{D}_m\mathbf{E}_m - \omega^2\epsilon\mathbf{E}_m &= i\omega\mathbf{J}_m && (113)\\
\mathcal{D}_{-m}\mathcal{D}_{-m}\mathbf{E}_{-m} - \omega^2\epsilon\mathbf{E}_{-m} &= i\omega\mathbf{J}_{-m} && (114)\\
\hat{\sigma}\mathcal{D}_{-m}\mathcal{D}_{-m}\hat{\sigma}\hat{\sigma}\mathbf{E}_{-m} - \omega^2\hat{\sigma}\epsilon\hat{\sigma}\hat{\sigma}\mathbf{E}_{-m} &= i\omega\hat{\sigma}\mathbf{J}_{-m} && (115)\\
\mathcal{D}_m\mathcal{D}_m(\hat{\sigma}\mathbf{E}_{-m}) - \omega^2\epsilon(\hat{\sigma}\mathbf{E}_{-m}) &= i\omega\mathbf{J}_m && (116)\\
\implies \quad \mathbf{E}_{-m} &= \hat{\sigma}\mathbf{E}_m && (117)
\end{aligned}$$

where

$$\mathcal{D}_m = \begin{pmatrix} 0 & -\frac{\partial}{\partial z} & \frac{im}{r} \\ \frac{\partial}{\partial z} & 0 & -\frac{\partial}{\partial r} \\ -\frac{im}{r} & \frac{1}{r}\frac{\partial}{\partial r}r & 0 \end{pmatrix},$$

Note we used Mathematica to check that $\hat{\sigma}\mathcal{D}_{-m}\mathcal{D}_{-m}\hat{\sigma} = \mathcal{D}_m\mathcal{D}_m$ (see Fig. 1). This is NOT rigorous because some of the differential operators don't commute but you should do the check by hand.
Unfortunately, this only holds true for the continuous axisymmetry operator and doesn't seem to carry over to the discrete axisymmetry operator:

$$\mathcal{D}_m^{\text{discrete}} = \begin{pmatrix} 0 & -\frac{\partial}{\partial z} & \frac{im}{r} + \frac{1}{r}\frac{\partial}{\partial\theta} \\ \frac{\partial}{\partial z} & 0 & -\frac{\partial}{\partial r} \\ -\frac{im}{r} - \frac{1}{r}\frac{\partial}{\partial\theta} & \frac{1}{r}\frac{\partial}{\partial r}r & 0 \end{pmatrix},$$

However, we expect that for small azimuthal periods (in the effective index limit), this property should approximately hold true.

```
In[27]:= D1 = {{0, -z, t}, {z, 0, -r}, {-t, rr, 0}};
         D1 // MatrixForm
         D2 = {{0, -z, -t}, {z, 0, -r}, {t, rr, 0}};
         D2 // MatrixForm
         S = {{1, 0, 0}, {0, -1, 0}, {0, 0, 1}};
         S // MatrixForm
         S.D2.D2.S // MatrixForm
         D1.D1 // MatrixForm
```

Out[28]//MatrixForm=

$$\begin{pmatrix} 0 & -z & t \\ z & 0 & -r \\ -t & rr & 0 \end{pmatrix}$$

Out[30]//MatrixForm=

$$\begin{pmatrix} 0 & -z & -t \\ z & 0 & -r \\ t & rr & 0 \end{pmatrix}$$

Out[32]//MatrixForm=

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Out[33]//MatrixForm=

$$\begin{pmatrix} -t^2 - z^2 & rr\,t & r\,z \\ r\,t & -r\,rr - z^2 & t\,z \\ rr\,z & t\,z & -r\,rr - t^2 \end{pmatrix}$$

Out[34]//MatrixForm=

$$\begin{pmatrix} -t^2 - z^2 & rr\,t & r\,z \\ r\,t & -r\,rr - z^2 & t\,z \\ rr\,z & t\,z & -r\,rr - t^2 \end{pmatrix}$$

Figure 1: Checking in Mathematica that $\hat{\sigma}\mathcal{D}_{-m}\mathcal{D}_{-m}\hat{\sigma} = \mathcal{D}_m\mathcal{D}_m$.

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